

Effects of Aerobic Exercise on Mild Cognitive Impairment

Objectives To examine the effects of aerobic exercise on cognition and other biomarkers associated with Alzheimer disease pathology for older adults with mild cognitive impairment, and assess the role of sex as a predictor of response. **Design** Six-month, randomized, controlled, clinical trial. **Setting** Veterans Affairs Puget Sound Health Care System clinical research unit. **Participants** Thirty-three adults (17 women) with amnesic mild cognitive impairment ranging in age from 55 to 85 years (mean age, 70 years). **Intervention** Participants were randomized either to a high-intensity aerobic exercise or stretching control group. The aerobic group exercised under the supervision of a fitness trainer at 75% to 85% of heart rate reserve for 45 to 60 min/d, 4 d/wk for 6 months. The control group carried out supervised stretching activities according to the same schedule but maintained their heart rate at or below 50% of their heart rate reserve. Before and after the study, glucometabolic and treadmill tests were performed and fat distribution was assessed using dual-energy x-ray absorptiometry. At baseline, month 3, and month 6, blood was collected for assay and cognitive tests were administered. **Main Outcome Measures** Performance measures on Symbol-Digit Modalities, Verbal Fluency, Stroop, Trails B, Task Switching, Story Recall, and List Learning. Fasting plasma levels of insulin, cortisol, brain-derived neurotrophic factor, insulinlike growth factor-I, and β -amyloids 40 and 42. **Results** Six months of high-intensity aerobic exercise had sex-specific effects on cognition, glucose metabolism, and hypothalamic-pituitary-adrenal axis and trophic activity despite comparable gains in cardiorespiratory fitness and body fat reduction. For women, aerobic exercise improved performance on multiple tests of executive function, increased glucose disposal during the metabolic clamp, and reduced fasting plasma levels of insulin, cortisol, and brain-derived neurotrophic factor. For men, aerobic exercise increased plasma levels of insulinlike growth factor I and had a favorable effect only on Trails B performance. **Conclusions** This study provides support, using rigorous controlled methodology, for a potent nonpharmacologic intervention that improves executive control processes for older women at high risk of cognitive decline. Moreover, our results suggest that a sex bias in cognitive response may relate

to sex-based differences in glucometabolic and hypothalamic-pituitary-adrenal axis responses to aerobic exercise.